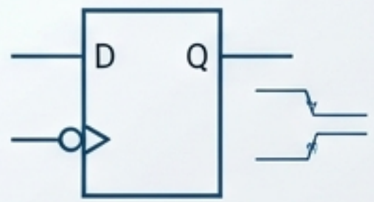
FPGA Design & Simulation: Sequential Logic Analysis

From Single-Bit Flip-Flops to Synchronous Counters

Context: Technical deep-dive based on the FPGA Design Exercise by Mahi P. Doshi.

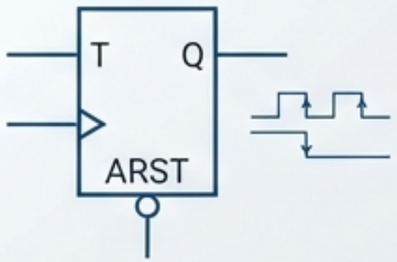
The Logic Ladder: From Storage to Sequencing

Capture



D Flip-Flop
(Negative Edge)

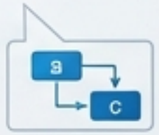
Control



T Flip-Flop
(Async Reset)

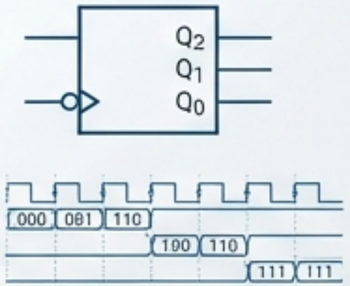
Order

```
always @(posedge clk) begin
  a = b + 1;
  c = a << 2;
end
```



Blocking
Assignments

Integration



Synchronous
3-Bit Counter

We validate correct hardware behavior by analyzing code alongside simulation waveforms, verifying that logic translates into reality.

Precision Data Capture: The Negative Edge D Flip-Flop

The Code

```
`timescale 1ns/1ps
module d_ff_neg (
    input clk,
    input d,
    output reg q
);
    always @(negedge clk)
        q <= d;
endmodule
```

The Reality

The ‘Photographer’ Circuit: Updates to output ‘q’ occur strictly on the falling edge of the clock. Stability is maintained between edges.

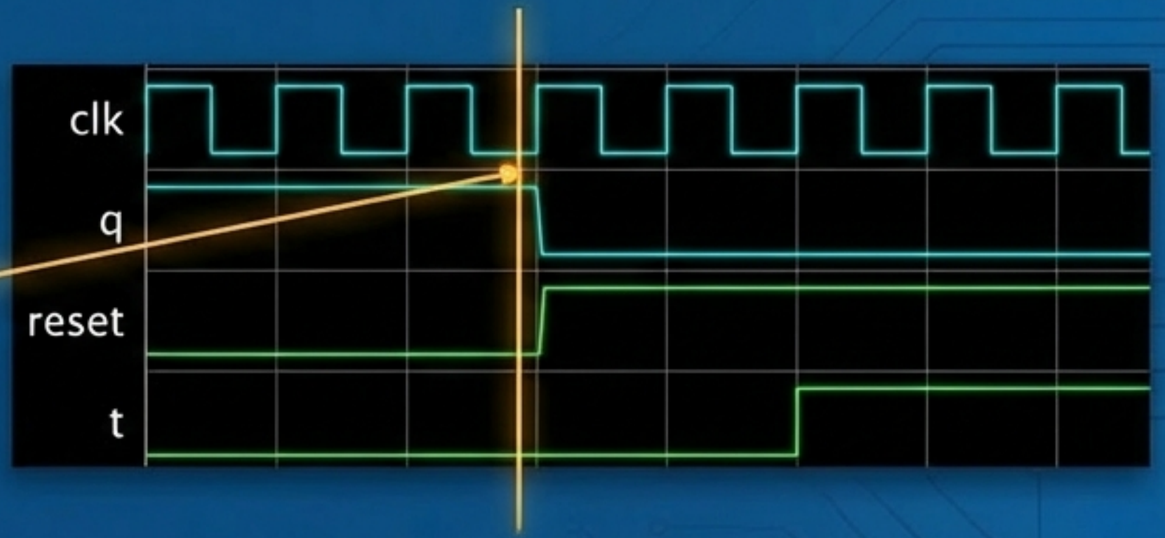
Asynchronous Control: The T Flip-Flop

The Code

```
module t_ff_async (
   input clk,
   input reset,
   input t,
   output reg q
);

always @(posedge clk or posedge reset)
begin
   if (reset)
      q <= 0;
   else if (t)
      q <= ~q;
end
endmodule
```

The Reality



Priority Logic: The asynchronous reset forces the output to 0 immediately, overriding the clock signal. This ensures the system can always be initialized to a known state.

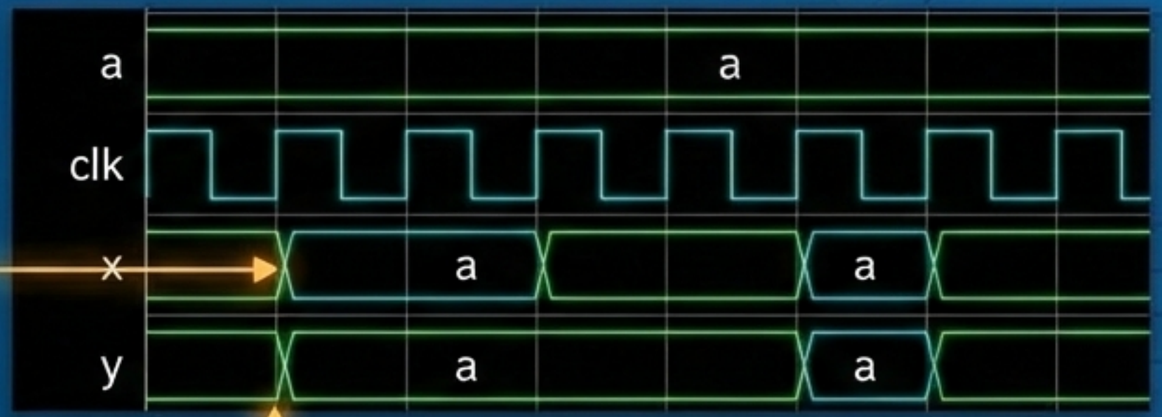
Execution Order: The Mechanics of Blocking Assignments

The Code

```
module blocking_example (
    input clk,
    input a,
    output reg x,
    output reg y
);

always @(posedge clk)
begin
    x = a;
    y = x;
end
endmodule
```

The Reality



Sequential Execution: In a blocking assignment (=), the second statement sees the newly updated value of x immediately. This mimics software-like progression within a single clock cycle.

System Integration: The 3-Bit Synchronous Up Counter

```
module up_counter (
  input clk,
  input reset,
  output reg [2:0] q
);

always @(posedge clk or posedge reset)
begin
  if (reset)
     q <= 3'b000;
  else
     q <= q + 1;
end
endmodule
```

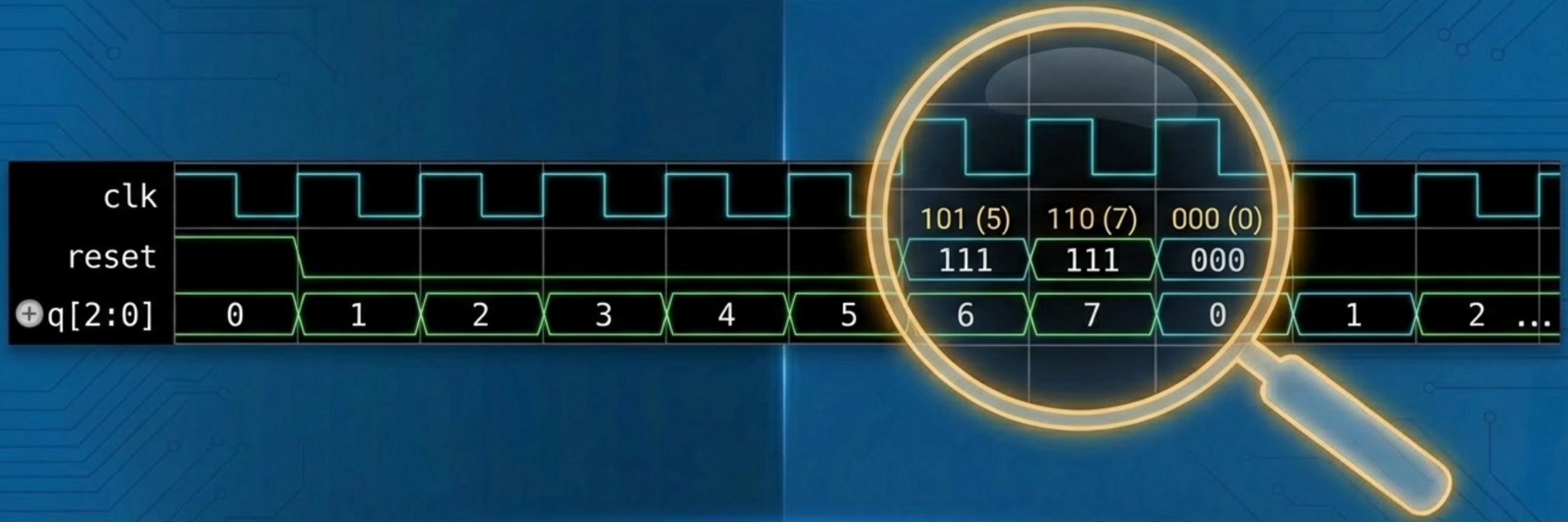
3-Bit Register Bus

Synchronous Trigger

Accumulation Logic

The Accumulator: All three bits update simultaneously on the rising clock edge. This minimizes propagation delay compared to asynchronous ripple counters.

Verification: Visualizing the Binary Sequence



Cycle Complete: The counter progresses linearly from 0 to 7. Upon reaching the maximum 3-bit value, it naturally rolls over to 0 on the next clock edge.

Component Architecture: Selecting the Right Tool

D Flip-Flop	T Flip-Flop	Synchronous Counter
Role: Data Capture. Best For: Timing-sensitive snapshots and pipeline stages.	Role: State Toggling. Best For: Frequency division and simple on/off switches.	Role: Sequencing. Best For: Event counting, addressing memory, and timing control.

Successful FPGA architecture relies on combining these primitives to manage state and time.

Engineer's Log: Timing Precision & Design Warnings

```
warning: Some design elements
have no explicit time unit and/or
time precision. This may cause
confusing timing results.
```

Critical Insight: The `timescale` Directive

While the source code defines `timescale 1ns/1ps`, the compiler warns of potential precision mismatches. In real-world hardware, undefined time units can lead to critical simulation failures. Explicitly defining time scales is mandatory for verification.

Final Validation

- Verified negative-edge data capture (D-FF)
- Validated asynchronous reset priority (T-FF)
- Demonstrated sequential execution flow (Blocking Assignment)
- Confirmed synchronous state accumulation (3-Bit Counter)

The logic holds. The design is ready for synthesis.